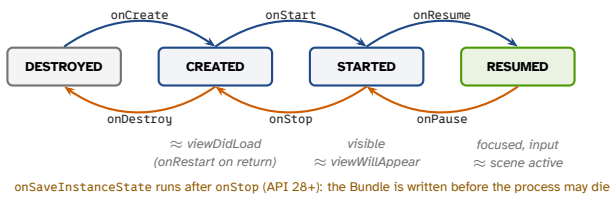
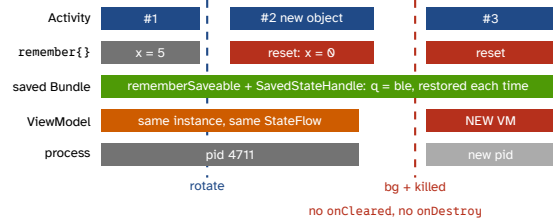


In one line: An Activity is **destroyed and recreated** on every configuration change (rotation, locale, dark mode, window size); the **ViewModel** survives that, the **saved-state Bundle** survives **process death**, and Compose redraws UI as a **function of state** — re-running only the composables that *read* a changed State.

Activity Lifecycle.State + callbacks



What survives what



Lifecycle, ViewModel, process death

- Config change $\Rightarrow$ onPause/onStop/onDestroy + a **new** Activity onCreate(savedInstanceState): fields and views gone. android:configChanges opts out — avoid.
- Fragment:** its *view* dies (onDestroyView) while the instance sits on the back stack $\Rightarrow$ observe with viewLifecycleOwner.
- ViewModel** lives in a ViewModelStore (Activity, Fragment or nav entry) carried across recreation; onCleared() only when the owner *finishes*. viewModelScope = SupervisorJob + Main.immediate, cancelled there.
- Process death:** killed in background; on return a new process rebuilds the Activity *from the Bundle* + a new VM. SavedStateHandle (VM) and rememberSaveable (UI) write that Bundle — ids, not data (Binder $\approx$ 1 MB $\rightarrow$ TransactionTooLargeException).
- Activity Context in a VM leaks it per rotation: applicationContext.

Compose — the model

- Phases: **composition** $\rightarrow$ **layout** $\rightarrow$ **draw**. Snapshot state records which scope **read** which State; a write re-runs only those scopes.
- remember survives recomposition only; remember(key) recomputes on key change; rememberSaveable also survives rotation + death (Bundle types or a Saver). **Hoisting:** stateless Foo(value, onValueChanged).
- Skipping:** skipped if all params are **stable** and equal. Stable: primitives, String, lambdas, val-only classes, @Immutable/@Stable. Unstable: var, List (interface), types from non-Compose modules. **Strong skipping** (default, Kotlin 2.0.20+): unstable params compared by identity, lambdas auto-remembered.
- Modifier order** wraps: size(100.dp).padding(10.dp) = 80 dp content; reversed = 120 dp total. clickable first = padding tappable.

Example — VM + hoisted screen

```
class SearchVM(saved: SavedStateHandle, val repo: Repo) : ViewModel() {
    val query = saved.getStateFlow("q", "") // survives death
    private val _ui = MutableStateFlow<Ui>(Ui.Idle)
    val ui = _ui.asStateFlow() // read-only
    fun onQuery(q: String) { saved["q"] = q
        viewModelScope.launch { _ui.value = repo.search(q) } }
}

@Composable fun Search(vm: SearchVM = viewModel()) {
    val q by vm.query.collectAsStateWithLifecycle()
    SearchBar(q, onChange = vm::onQuery) // hoisted
    LaunchedEffect(q) { delay(300); log(q) } // per q
}
```

Compose $\leftrightarrow$ SwiftUI (incl. effects)

Compose / Android	SwiftUI / iOS
@Composable fun — re-run function	body — value tree, diffed
remember { mutableStateOf() }	@State
hoisted value + onValueChanged	@Binding
rememberSaveable	@SceneStorage
ViewModel + StateFlow, collectAsStateWithLifecycle	@Observable model in @State
CompositionLocal	@Environment
Modifier chain · slot lambdas	modifiers : @ViewBuilder
items(list, key = { it.id }) (else index)	ForEach(id:)
AndroidView · ComposeView	UIViewRepresentable · UIHostingController · NavigationStack(path:)
NavHost + back stack	
LaunchedEffect(k): coroutine on enter, relaunched when k changes, cancelled on leave	.task(id:)
DisposableEffect(k): setup + onDispose{ }	onAppear / onDisappear
SideEffect{ }: after every successful recomposition, push to non-Compose code	updateUIView
rememberCoroutineScope: coroutines from callbacks	Task { } in an action
rememberUpdatedState: latest value in a long effect, no restart	—
derivedStateOf: recompose only when the result changes	computed property
snapshotFlow(State $\rightarrow$ Flow) · produceState	—

Traps

- Rotation recreates the Activity** — the iOS reflex “the VC stays” loses state. Three tiers: remember < ViewModel < Bundle.
- VM does not survive process death.** “Don’t keep activities” only destroys Activities; real death: adb shell am kill <pkg> while backgrounded.
- mutableStateOf(0)** without remember resets each recomposition. LaunchedEffect(Unit) never restarts (stale id).
- Fast state (scrollState.value)** read high up recomposes the screen — defer the read: lambda, graphicsLayer{ }, derivedStateOf.
- Bodies re-run often, in any order:** no I/O or writes.

Remember

remember < ViewModel < Bundle · state down, events up · read late, key your effects.